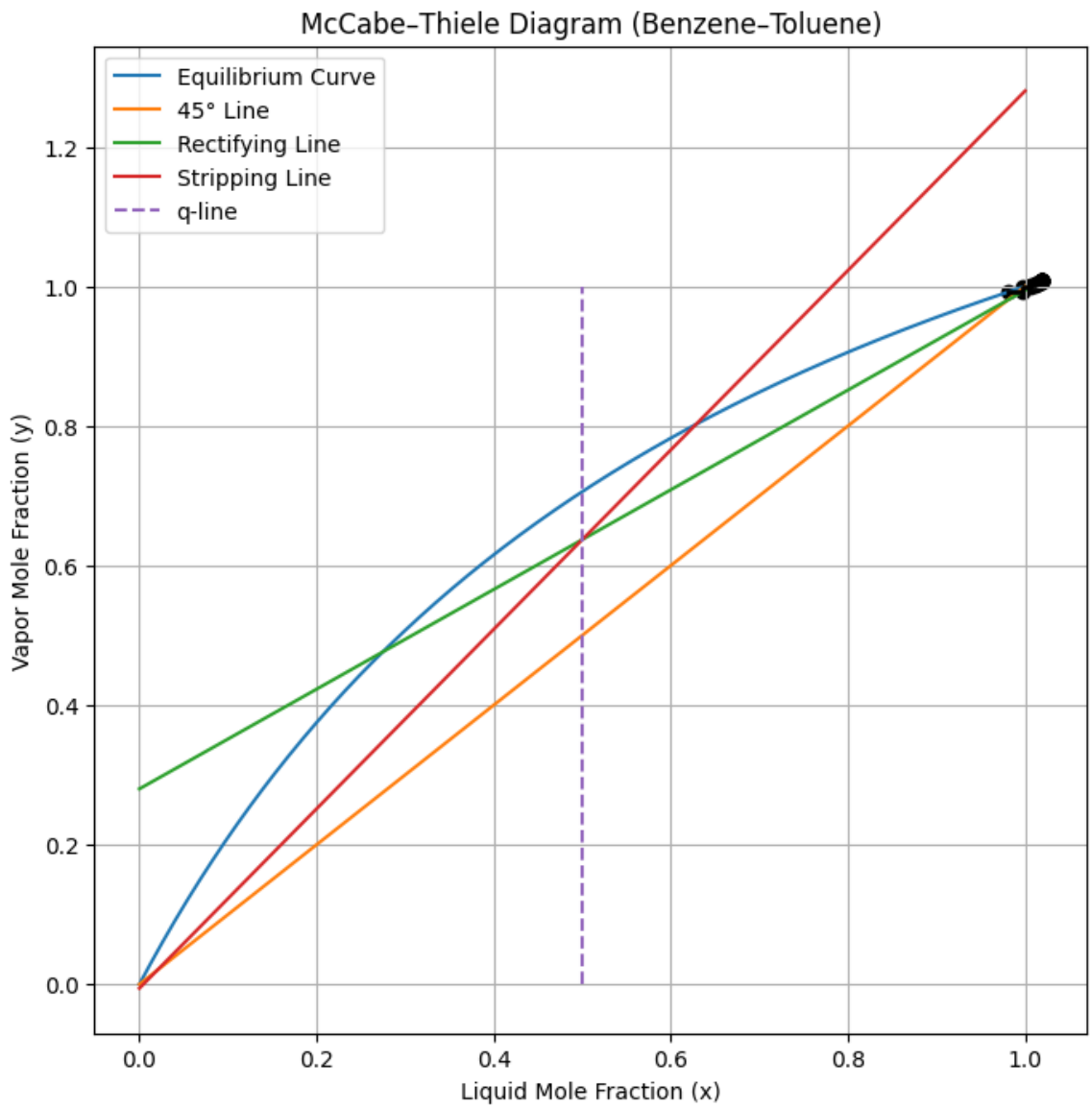
Project Roadmap: AI-Based Distillation Column Optimization

1. McCabe–Thiele Plot and Using it as a Baseline

Purpose:
The McCabe–Thiele graphical method was used as the initial theoretical design step for the distillation column.
This method provides an approximate estimate of the number of theoretical stages required for separation.

Strategy Behind This Step:
• Vapor–liquid equilibrium relations were defined assuming constant relative volatility for the benzene–toluene system.
• The equilibrium curve and the 45° line were plotted.
• Rectifying and stripping operating lines were constructed based on reflux ratio and product compositions.
• A q-line was added to represent the feed condition.
• Stage stepping between the equilibrium curve and operating lines was performed to estimate the number of theoretical stages.

Why This Step Was Important:
This theoretical design served as a baseline reference before rigorous simulation and optimization.



2. Rigorous Distillation Simulation in DWSIM

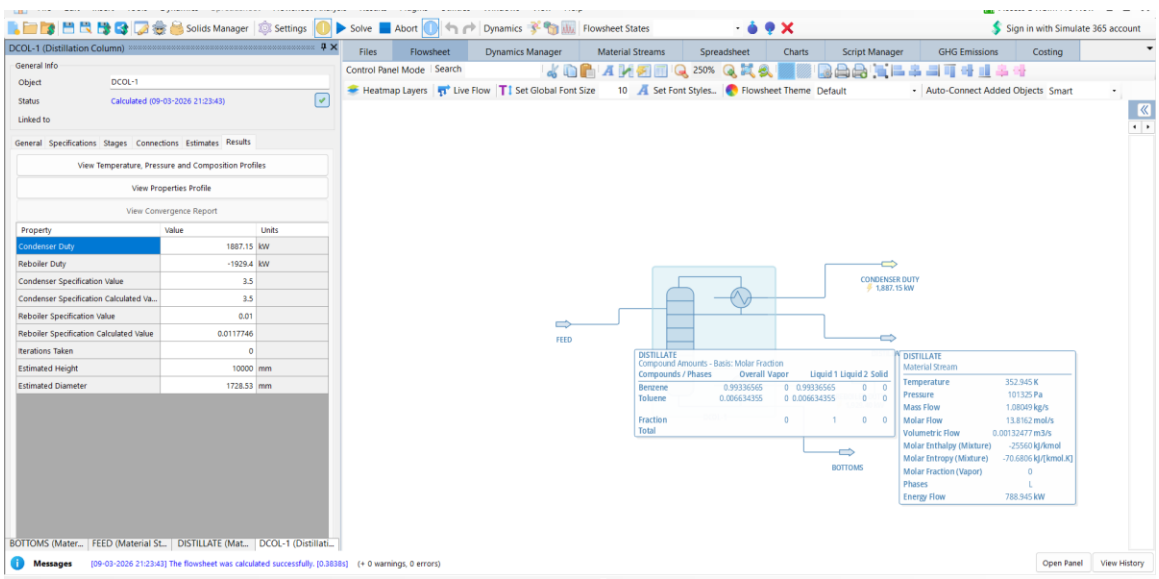
Purpose:
After obtaining the theoretical estimate, a rigorous simulation of the distillation column was performed using DWSIM.

Steps Performed:

• Components benzene and toluene were selected.
• An appropriate thermodynamic model was chosen.
• A distillation column was inserted into the flowsheet.
• Feed stream conditions were specified (composition, temperature, pressure, flow rate).
• Condenser and reboiler specifications were defined.
• The number of stages and reflux ratio were set based on theoretical estimates.
• The simulation was executed until convergence was achieved.

Why This Step Was Important:
Rigorous simulation accounts for detailed thermodynamics, mass balances, and energy balances.
It provides a more accurate prediction of distillation performance compared to the simplified McCabe–Thiele method.

Initial Simulation:



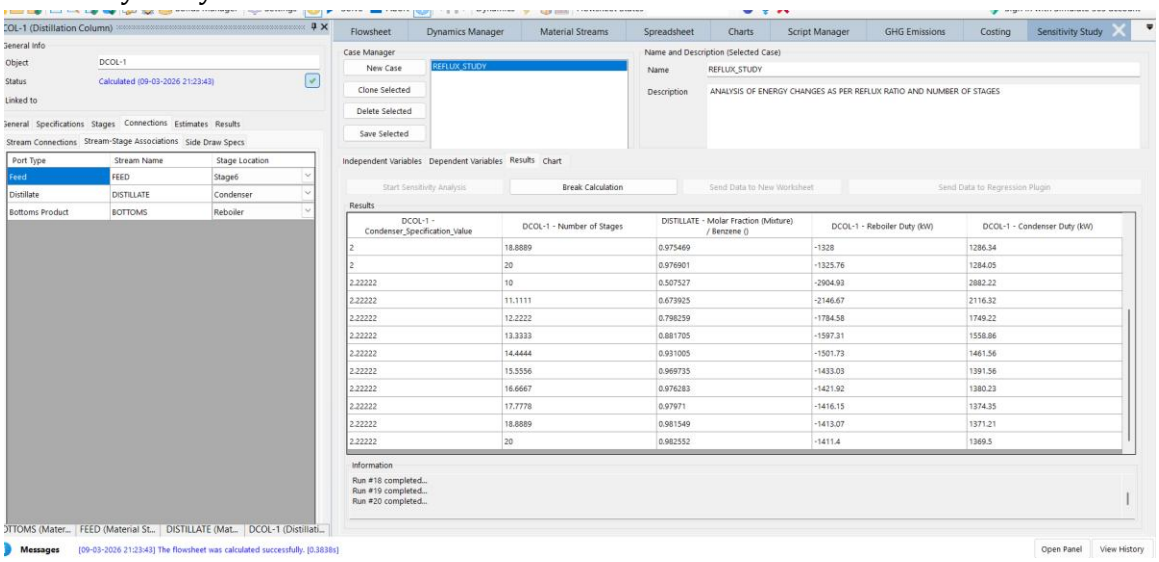
3. Dataset Generation Using Sensitivity Analysis

Purpose:
To train the machine learning model, a dataset representing different operating conditions was required.

Steps Performed:
• DWSIM sensitivity analysis was used to vary key design variables automatically.
• Reflux ratio and number of stages were selected as independent variables.
• Distillate purity and reboiler duty were recorded as dependent variables.
• Multiple simulations were generated automatically to create a structured dataset.
• The results were exported to an Excel file.

Why This Step Was Important:
The generated dataset served as the training data for the machine learning surrogate model.

Sensitivity Analysis:



4. Coding and Training the AI Model

Purpose:
Machine learning models were developed to learn the relationship between column operating variables
and performance metrics.

Steps Performed:
- The dataset was imported into Python.
- Data preprocessing and cleaning were performed.
- Input variables were defined (reflux ratio, number of stages).
- Output variables were defined (purity, energy consumption).
- Machine learning regression models were trained.
- **Bayesian optimization** was implemented to search for optimal operating conditions.

Why This Step Was Important:
The surrogate model allows fast prediction of column performance without running new simulations.

Loading the Datasets:

```
from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]

data = pd.read_excel(filename)

print("Dataset Loaded\n")

print(data.head())
print("\nColumns:")
print(data.columns)
print("\nShape:",data.shape)
```

Training the Models:

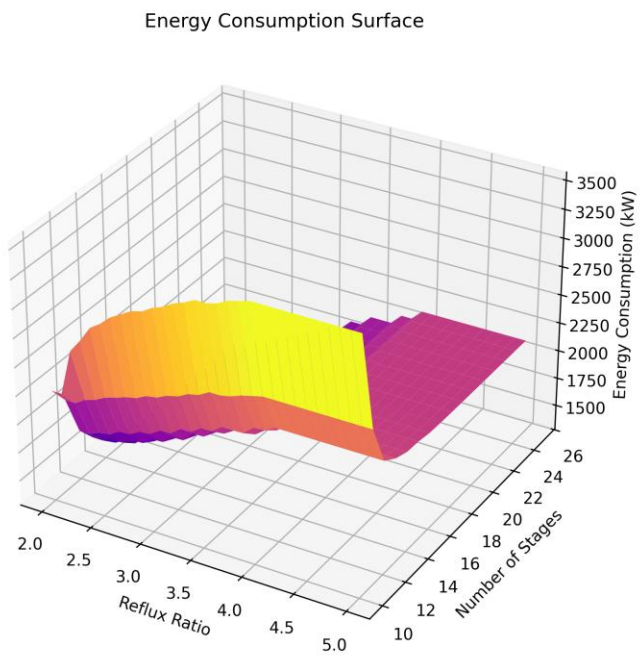
```
[25]
1s
from sklearn.ensemble import RandomForestRegressor


purity_model = RandomForestRegressor(
    n_estimators=200,
    random_state=42
)

purity_model.fit(X, y_purity)



energy_model = RandomForestRegressor(
    n_estimators=200,
    random_state=42
)

energy_model.fit(X, y_energy)


print("Models trained successfully")
```

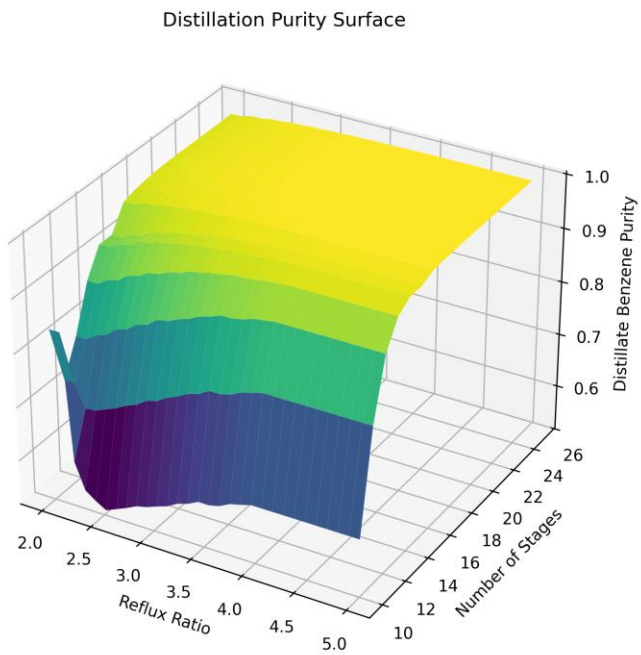
```
Models trained successfully
```

Bayesian Optimzation:

```
from skopt import gp_minimize
from skopt.space import Real, Integer


search_space = [

    Real(2,5,name='reflux'),
    Integer(10,25,name='stages')

]


result = gp_minimize(

    objective,
    search_space,
    n_calls=40,
    random_state=42

)


opt_reflux = result.x[0]
opt_stages = result.x[1]

opt_purity = purity_model.predict([[opt_reflux,opt_stages]])[0]
opt_energy = energy_model.predict([[opt_reflux,opt_stages]])[0]
```

Generated Purity and Energy Surface plots:





5. Obtaining Results and Validation Using DWSIM

Purpose:
The optimal operating conditions predicted by the machine learning model were validated through rigorous simulation.

Steps Performed:
• The optimized reflux ratio and number of stages predicted by the AI model were identified.
• These parameters were applied to the DWSIM distillation column.
• The simulation was executed again.
• Predicted purity and energy values were compared with simulation results.

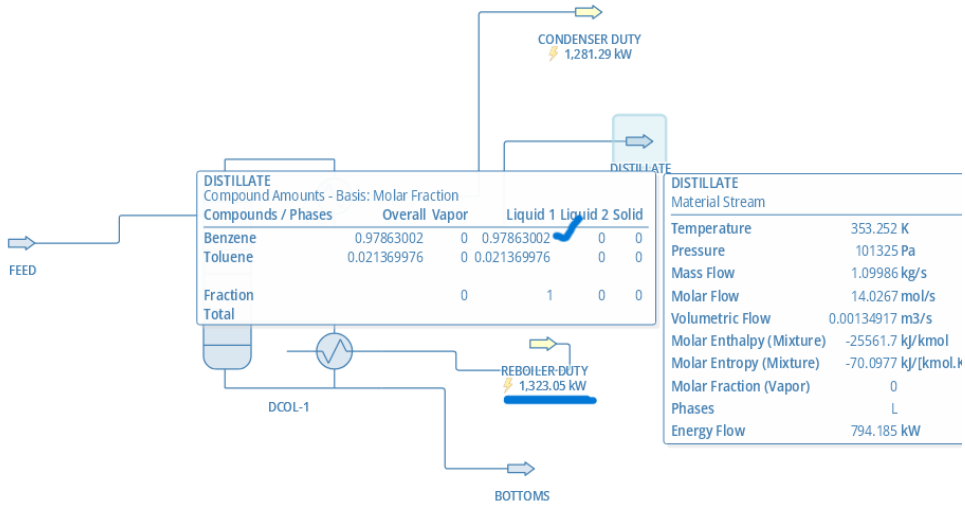
Additional Analysis Performed:
• A comparison graph was created showing stage predictions from McCabe–Thiele, DWSIM, and AI optimization.
• 3D surface plots were generated to visualize how reflux ratio and number of stages influence purity and energy consumption.

Why This Step Was Important:
Validation confirmed that the AI-based optimization approach could accurately identify improved operating conditions.

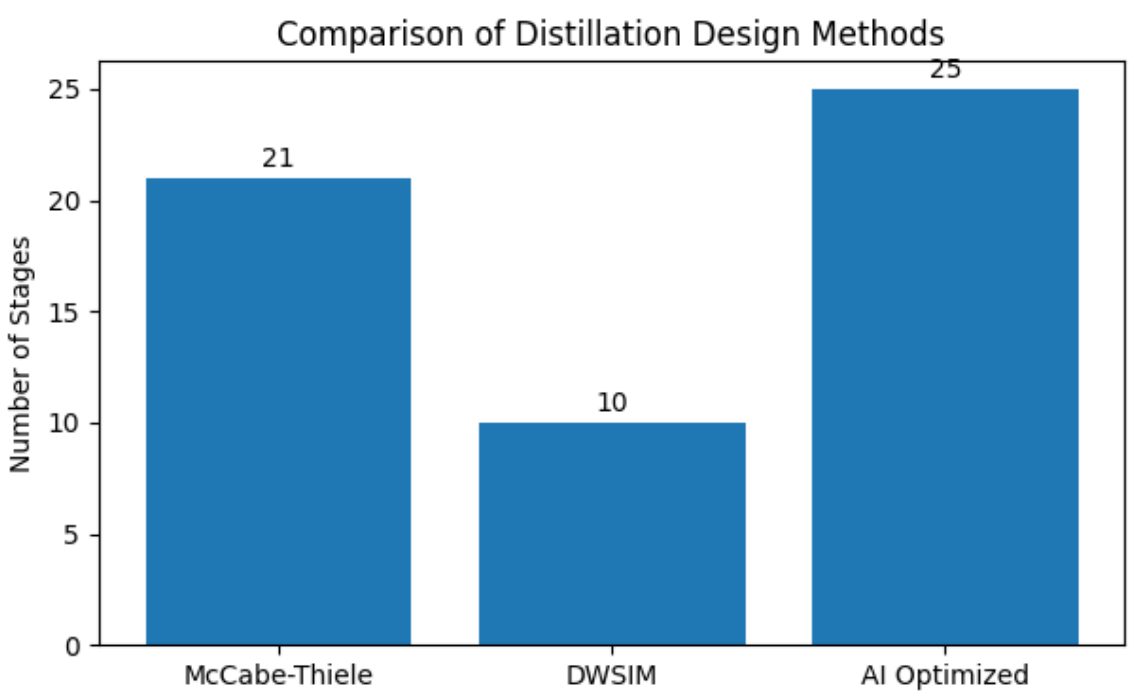
AI Optimized results:

```
...
    ==============================
     AI OPTIMIZATION RESULTS
    ==============================

    Optimal Reflux Ratio  : 2.000
    Optimal Stages        : 25
    Predicted Purity      : 0.9764
    Predicted Energy (kW) : 1329.25

    Files saved:
     - purity_surface.png
     - energy_surface.png
     - optimal_conditions.csv
     - optimization_surface.csv
```

Validation of the results by simulation in DWSIM:



Comparison of various Designs:



THANK YOU!

As a First Year B.Tech Student making this project successful was huge for me!
A random idea turned into something meaningful.
I spent 3 days learning deep theory of the concepts of distillation and I can say that I have mastered the topic at a theoretical level from various online resources.
The next 5 days I spent in simulation after failing multiple times in convergence errors, and finally on the last day I wrote the optimization code.
There may be multiple errors in the project , and I am quite sure I will rectify them with constructive discussions!